- Most points of a uniformly distributed dataset are not close to the center of mass, but far away towards the surface of the cluster;
- If data is uniformly distributed in a hypercube (a domain where each feature is uniformly distributed in a fixed range), then in high dimensions most points are close to the faces of the hypercube:
  - Approximating a hypercube with its *subscribed hypersphere* will leave out most of the points;
  - To include all the points, we need the hypersphere **superscribed** to the hypercube, and – as we have seen in chapter 10, section 10.5.1, the amount of volume wasted grows exponentially with the number of dimensions;
  - In a `d`-dimensional space, with `d >> 10`, under certain reasonable assumptions for data distribution, the nearest neighbor problem becomes ill-defined[8], because in high dimensions the ratio between the distance from a target to the nearest and farthest neighbor becomes almost 1; for instance, if points are equally spaced (like placed on a grid), then the `2d` closest points to any centroid are all at the same distance from it: this means that finding the nearest neighbor of a point becomes challenging.

In simpler words, for higher-dimensional datasets, unless the distribution of clusters is exactly spherical, the spheres needed to include all points are so large that they will likely overlap each other for a significant portion of their volume; moreover, for close-to-uniformly distributed datasets and when many centroids are used, the search for the nearest